ModelIt!

Student Activity Pack

Why Is Lithium the New Gold?

Modeling the Resource Economics of the Electric Vehicle Revolution

Grade 10L2 | NGSS HS-ESS3-1, HS-PS1-3 | Cause and Effect / Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you know about where the materials in your phone battery come from?

2 Why do you think electric vehicle prices have been so volatile in recent years?

3 Draw a diagram showing how you think lithium gets from the ground into a car battery.

4 What trade-offs might exist between fighting climate change and mining for battery materials?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Lithium Demand | Mining Rate | Environmental Damage | Battery Performance | Recycling Rate | Market Price | EV Adoption

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Steady Growth

Moderate EV adoption, current mining capacity

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: EV Boom, No Recycling

Rapid EV adoption, zero recycling

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Circular Economy

High recycling, moderate adoption

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Geopolitical Risk — The concentration of lithium reserves in a few countries (Australia, Chile, Argentina, China) creates supply vulnerability to political instability, trade disputes, and export restrictions

Alternative Chemistry — Research into sodium-ion, solid-state, and lithium-sulfur batteries that could reduce or eliminate lithium dependence in future battery generations

Water Scarcity Index — The ratio of lithium extraction water consumption to available freshwater in mining regions, which determines whether extraction is sustainable or destroys local water supplies

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why has a soft, silvery metal that most people have never heard of become the most fought-over resource on the planet? Use evidence from your simulation to support your answers.

1. **How does your model reveal the tension between the urgency of climate action and the reality of resource extraction timelines?**

2. **What feedback loops in your model create the most instability in lithium markets?**

3. **How would your model's predictions change if a breakthrough in sodium-ion batteries eliminated lithium demand for EVs?**

4. **What does your model suggest about the difference between a linear economy and a circular economy for critical minerals?**

5. **What ethical considerations does your model raise about who bears the environmental cost of the green energy transition?**

8 STEM Challenge: Design a Lithium Supply Chain Resilience Plan

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YOUR ENGINEERING MISSION

Design a 20-year lithium supply strategy for an EV manufacturer that balances cost, environmental responsibility, and supply security.

THE SCENARIO

A major electric vehicle company projects it will need 10x its current lithium supply within eight years. Current suppliers cannot guarantee delivery. Your team has been hired to design a comprehensive supply strategy that ensures the company can meet production targets without contributing to environmental disaster.

Engineering Design Thinking

1. WHAT MIX OF LITHIUM SOURCES — BRINE, HARD-ROCK, RECYCLED, AND NEXT-GENERATION — WOULD YOU RECOMMEND?
2. HOW WOULD YOU STRUCTURE LONG-TERM SUPPLY CONTRACTS TO HEDGE AGAINST PRICE SWINGS?
3. WHAT ROLE SHOULD BATTERY CHEMISTRY RESEARCH PLAY IN REDUCING LITHIUM DEPENDENCE?
4. HOW WOULD YOU COMMUNICATE YOUR ENVIRONMENTAL STRATEGY TO CONSUMERS AND INVESTORS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Supply Chain Engineers and Resource Economists analyze global material flows, predict shortages, and design resilient procurement strategies. They work for automotive companies, mining firms, government agencies, and consulting firms, earning $75,000–$150,000/year. Battery Materials Scientists developing next-generation chemistries earn $90,000–$160,000/year.